DIVISIBILITY AND RATIONAL ROOTS OF INTEGER POLYNOMIALS

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Abstract. We give a short, self-contained account of the elementary divisibility properties of polynomials with integer coefficients. The main result is the classical fact that every rational root of a monic integer polynomial is an integer dividing the constant term. Along the way we record several consequences of the basic congruence $a - b \mid P(a) - P(b)$, including a criterion for the absence of integer roots and a remark on periodic points.

1. Introduction

Polynomials with integer coefficients enjoy a rigid arithmetic structure: their values at integers are constrained by divisibility relations that do not hold for arbitrary real polynomials. The most basic of these is the observation that $a - b$ divides $P(a) - P(b)$ whenever $a, b \in \mathbb{Z}$ and $P \in \mathbb{Z}[x]$. Combined with unique factorisation in $\mathbb{Z}$, it yields the *rational root theorem*, which in turn shows that the rational roots of a monic integer polynomial are integers.

These facts are standard and appear in many textbooks, but they are usually scattered over several chapters. The purpose of this note is to collect them in one place, with complete proofs, in a form convenient for teaching and for quick reference. Our main result is Theorem 2.9. In Section 3 we combine it with Eisenstein's criterion and derive two simple non-existence results for rational roots.

2. Definitions and main results

Definition 2.1. Throughout, $P \in \mathbb{Z}[x]$ denotes a nonzero polynomial, written $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ with $a_i \in \mathbb{Z}$ and $a_n \neq 0$, so that $n = \deg P$. We call a_n the *leading coefficient* and $a_0 = P(0)$ the *constant term*; P is *monic* if $a_n = 1$. A *root* of P is a number r with $P(r) = 0$. A rational number r is *written in lowest terms* as $r = p/q$ if $p, q \in \mathbb{Z}$, $q \geq 1$ and $\gcd(p, q) = 1$; every $r \in \mathbb{Q}$ has exactly one such representation.

Lemma 2.2. *Let $P \in \mathbb{Z}[x]$ and $a, b \in \mathbb{Z}$. Then $a - b$ divides $P(a) - P(b)$.*

Proof. For every $k \geq 1$ we have the factorisation

$$a^k - b^k = (a - b) \sum_{j=0}^{k-1} a^j b^{k-j},$$

so $a-b \mid a^k-b^k$. Since $P(a)-P(b)=\sum_{k=1}^{n} a_k\,(a^k-b^k)$, the claim follows by linearity. □

Corollary 2.3. *Let $m \geq 1$ and let $P \in \mathbb{Z}[x]$ be monic. If $a, b \in \mathbb{Z}$ satisfy $a \equiv b \pmod{m}$, then $P(a) \equiv P(b) \pmod{m}$.*

Proof. By hypothesis $m \mid a-b$, and by Lemma 2.2 $a-b \mid P(a)-P(b)$. Hence $m \mid P(a)-P(b)$ by transitivity of divisibility. □

Lemma 2.4. *Let $m \geq 2$ and $P \in \mathbb{Z}[x]$. If $m \nmid P(k)$ for every $k \in \{1, 2, \ldots, m-1\}$, then P has no integer root.*

Proof. Suppose $r \in \mathbb{Z}$ satisfies $P(r)=0$, and let $k \in \{1, \ldots, m-1\}$ be the residue of r modulo m. Then $m \mid r-k$, and by Lemma 2.2, $r-k \mid P(r)-P(k)=-P(k)$. Thus $m \mid P(k)$, contrary to the hypothesis. □

Lemma 2.5. *Let $P \in \mathbb{Z}[x]$ and $a, b \in \mathbb{Z}$. If $|P(a)-P(b)|=1$, then $|a-b|=1$.*

Proof. By Lemma 2.2, $a-b$ divides $P(a)-P(b)=\pm 1$, so $|a-b| \leq 1$. If $a=b$ then $P(a)-P(b)=0$, which is impossible; hence $|a-b|=1$. □

Lemma 2.6. *Let $P \in \mathbb{Z}[x]$. Every integer root r of P divides the constant term a_0; in particular $|r| \leq |a_0|$.*

Proof. Apply Lemma 2.2 with $a=r$ and $b=0$: $r \mid P(r)-P(0)=-a_0$. A nonzero divisor of an integer is at most that integer in absolute value. □

Proposition 2.7. *Let $P \in \mathbb{Z}[x]$ and let a, b, c be pairwise distinct integers with $P(a)=b$, $P(b)=c$ and $P(c)=a$. Then $|a-b|=|b-c|=|c-a|$.*

Proof. By Lemma 2.2, $a-b \mid P(a)-P(b)=b-c$, $b-c \mid P(b)-P(c)=c-a$ and $c-a \mid P(c)-P(a)=a-b$. All three differences are nonzero, so $|a-b| \leq |b-c| \leq |c-a| \leq |a-b|$, and equality holds throughout. □

Theorem 2.8 (Rational root theorem)**.** *Let $P \in \mathbb{Z}[x]$ have degree $n \geq 1$, and let $r=p/q \in \mathbb{Q}$, written in lowest terms, be a root of P. Then $p \mid a_0$ and $q \mid a_n$.*

Proof. Multiplying $P(p/q)=0$ by q^n gives $\sum_{i=0}^{n} a_i\, p^i q^{n-i}=0$. Every term with $i \geq 1$ is divisible by p, hence $p \mid a_0 q^n$; since $\gcd(p, q^n)=1$ we get $p \mid a_0$. Similarly every term with $i \leq n-1$ is divisible by q, hence $q \mid a_n p^n$, and $\gcd(q, p^n)=1$ gives $q \mid a_n$. □

Theorem 2.9. *Let $P \in \mathbb{Z}[x]$ be monic of degree $n \geq 1$, and let $r \in \mathbb{Q}$ be a root of P. Then $r \in \mathbb{Z}$ and r divides the constant term $P(0)$.*

Proof. Write $r=p/q$ in lowest terms. By Theorem 2.8, $q \mid a_n=1$, and since $q \geq 1$ this forces $q=1$, i.e. $r=p \in \mathbb{Z}$. Theorem 2.8 also gives $p \mid a_0=P(0)$. □

3. Applications

We recall a classical irreducibility criterion.

Theorem 3.1 (Eisenstein's criterion [1, Ch. IV, Thm. 3.1])**.** *Let $P = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ with $n \geq 1$, and let p be a prime such that $p \nmid a_n$, $p \mid a_i$ for all $0 \leq i < n$, and $p^2 \nmid a_0$. Then P is irreducible in $\mathbb{Z}[x]$.*

For instance, for every prime p and every $n \geq 1$ the polynomial $x^n - p$ is irreducible. The absence of rational roots, which is all we need below, can also be read off directly from Theorem 2.9.

Corollary 3.2. *Let p be a prime and $n \geq 2$. Then $x^n - p$ has no rational root.*

Proof. By Theorem 2.9, a rational root r would be an integer dividing p, so $r \in \{\pm 1, \pm p\}$. But then $|r|^n \in \{1, p^n\}$, and neither equals p because $n \geq 2$. □

Corollary 3.3. *Let $P \in \mathbb{Z}[x]$ be monic of degree $n \geq 1$ with $P(0)$ and $P(1)$ both odd. Then P has no rational root.*

Proof. By Theorem 2.9, any rational root of P is an integer, and by Lemma 2.4 with $m = 2$, P has no integer root. □

References

[1] S. Lang, *Algebra*, revised third edition, Graduate Texts in Mathematics 211, Springer, New York, 2002.

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